

Assignment No 09

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Question:

Design and implement an interface for Vehicles to represent common functionalities. Consider vehicles such as Bicycle, Bike, and Car, which share the following functionalities:

- **gearChange()**
- **speedUp()**
- **applyBrakes()**

An interface Vehicle that declares these methods. Implement the interface in the classes Bicycle, Bike, and Car, providing custom implementations for each method according to the specific vehicle type.

Code:

```
/*
PRBM STMT:
Design and implement an interface for Vehicles to represent common functionalities.

Vehicles:
- Bicycle
- Bike
- Car

Common Functionalities:
- gearChange()
- speedUp()
- applyBrakes()

Create an interface Vehicle and implement it in all classes.
*/

#include<iostream>
using namespace std;

class Vehicle{
public:
    virtual void gearChange() = 0;
    virtual void speedUp() = 0;
    virtual void applyBrakes() = 0;

    virtual ~Vehicle(){}
};

class Bicycle : public Vehicle{
public:
    void gearChange() override{
        cout << "Bicycle gear changed manually.\n";
    }

    void speedUp() override{
        cout << "Bicycle speed increased by pedaling faster.\n";
    }

    void applyBrakes() override{
        cout << "Bicycle brakes applied using hand brakes.\n";
    }
};

class Bike : public Vehicle{
public:
    void gearChange() override{
        cout << "Bike gear changed using gearbox.\n";
    }

    void speedUp() override{
```

```

        cout << "Bike speed increased using accelerator.\n";
    }

    void applyBrakes() override{
        cout << "Bike brakes applied using disc brakes.\n";
    }
};

class Car : public Vehicle{
public:
    void gearChange() override{
        cout << "Car gear changed using transmission system.\n";
    }

    void speedUp() override{
        cout << "Car speed increased by pressing accelerator.\n";
    }

    void applyBrakes() override{
        cout << "Car brakes applied using brake pedal.\n";
    }
};

int main(){
    int choice;

    cout << "--- Vehicle Interface Program ---\n";
    cout << "1. Bicycle\n";
    cout << "2. Bike\n";
    cout << "3. Car\n";
    cout << "Enter choice: ";
    cin >> choice;

    Vehicle* v = nullptr;

    switch(choice){
        case 1:
            v = new Bicycle();
            break;

        case 2:
            v = new Bike();
            break;

        case 3:
            v = new Car();
            break;

        default:
            cout << "Invalid Choice\n";
            return 0;
    }

    cout << "\nVehicle Operations:\n";

    v->gearChange();
    v->speedUp();
    v->applyBrakes();

    delete v;

    return 0;
}

```

Output:

```
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 9> .\a.exe
--- Vehicle Interface Program ---
1. Bicycle
2. Bike
3. Car
Enter choice: 1

Vehicle Operations:
Bicycle gear changed manually.
Bicycle speed increased by pedaling faster.
Bicycle brakes applied using hand brakes.
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 9> .\a.exe
--- Vehicle Interface Program ---
1. Bicycle
2. Bike
3. Car
Enter choice: 2

Vehicle Operations:
Bike gear changed using gearbox.
Bike speed increased using accelerator.
Bike brakes applied using disc brakes.
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 9> .\a.exe
--- Vehicle Interface Program ---
1. Bicycle
2. Bike
3. Car
Enter choice: 3

Vehicle Operations:
Car gear changed using transmission system.
Car speed increased by pressing accelerator.
Car brakes applied using brake pedal.
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 9> █
```